

ESOF-4011: Applied Computational Intelligence Project

Objective: This project aims to provide students with an opportunity to design and implement an end-to-end computational intelligent agent to solve a practical problem, demonstrating their understanding of computational intelligent algorithms, teamwork, the application of engineering practices, and problem-solving skills.

Project type: Group project consisting of a maximum of two members.

Deliverables:

1- Logistics

- a. Each group will share a folder on Google Drive to show their implementation. The folder must contain at least "THREE" working models, data, and a document explaining how to download, configure, train, and test the models for the chosen problem.
- b. A code without documentation WILL NOT be accepted. The code must be written in a modular fashion. The code and intermediate reports (see next page) must also be uploaded to the D2L.

2- Final report

- a. A 5 to 8-page formal project report following the IEEE two-column template must be submitted. Microsoft Word and Overleaf version templates are available at <https://www.ieee.org/conferences/publishing/templates.html>.

Infrastructure:

1- Computing platform: It is recommended to use the [Google Colab cloud platform](#).

2- The software platform

- a. Python and its key libraries, like NumPy and Scikit-Learn
- b. Deep learning tools, such as PyTorch, Keras, or TensorFlow

Project Ideas: The following are some project ideas.

	Timetable Scheduling using Genetic Algorithms: Solve course or exam scheduling as an optimization problem.
	Reinforcement Learning for Autonomous Navigation: Train an agent to navigate a grid or simulated environment. For example, AWS DeepRacer . Read this Documentation .
	Predictive Maintenance: Predict equipment behaviour or failure (detect anomalies in sensor feeds) from historical data of industrial equipment. For example, Deep neural network modeling for accurate electric motor temperature prediction , Global attention-based encoder-decoder lstm

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	model for temperature prediction of permanent magnet synchronous motors.
	Drone-based Parking Permit Detection and Unauthorized Vehicle Recognition: Unauthorized parking is a persistent issue in large campuses and urban areas. This project aims to design an intelligent drone-based system that automatically detects parked vehicles, recognizes parking permits (or license plates), and identifies unauthorized vehicles using computational intelligence techniques.
	Students can also propose their own ideas; however, they must get approval from the course instructor before proceeding.

Deadline for Submission: Refer to the table below.

Table I: Detailed deliverables with expected due dates

Project Stage	Due date (By the end of)	20% total
1. Project proposal: Describe the problem, identify the input and target variables, and elaborate on the types of attributes (categorical, numerical, text, integer, real), the number of samples in the training set, and the test set.	3 rd week	2%
2. Data curation and conditioning: - Data preprocessing, data transformation, normalization, augmentation, etc. - You must explain why you chose these preprocessing methods. - How do these methods suit the problem and the raw data?	5 th week	3%
3. Model selection and sanity tests: - Research on plausible models (SVM, multilayer neural network, convolutional neural network, graph neural network, federated learning models, ensemble learning models) for the problem at hand. - Complete a sanity test on data preprocessing and hyperparameter settings. - Show the analysis and sanity test using figures and tables.	7 th week	5%
4. Final model: Finalizing the best analytical pipeline, feature engineering, model complexity (memory footprint, number of layers,	10 th week	5%

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number of trainable parameters, inference time, FLOPs, etc.), and training strategy (choice of the objective functions, learning rate, batch size, handling overfitting issues, evaluation metrics, etc.). Note: <i>Finalizing a model is not just purely influenced by the number of layers and the scores computed using evaluation metric(s). The model complexity (the number of trainable parameters, and processing time or FLOPS) also plays a key role. If we do not consider the model complexity, we may end up selecting a sub-optimal model.</i> <i>For instance, an LSTM-based regressor and 1-D CNN-based regressor may achieve the same R2 scores and MAE scores are very similar (say CNN has ~2% more error). But the CNN-based model is less complex than the LSTM model.</i> <i>So, to pick the optimal model, you should also analyze their total number of trainable parameters as well as their FLOPS (the total number of floating-point operations required for a single forward pass). The higher the FLOPs, the slower the model (highly complex architecture) and hence low throughput.</i> <i>There are many ways to compute the FLOPS, like this one Calculate Flops in Tensorflow and Pytorch are not equal? - General Discussion - TensorFlow Forum.</i> ■ Describing the framework with flowcharts, block diagrams, and mathematical expressions. ■ Appropriate figures, graphs, and tables should be incorporated.		
5. Project completion: ■ Model regularization and validation. ■ Several regularization and validation metrics must be implemented and compared with existing works. ■ You must explain the regularization methods. ■ 8 to 10-minute presentation and final report submission (if time permits)	March 30	5%

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Group dynamics and progress: The workload must be equally divided among group members (two students per group). Each student is expected to be fully conversant with all aspects of the project, including components for which they are not directly responsible.

Important Notes:

- Students must upload/share the deliverables by the deadline. Students can still upload their deliverables **within 48 hours from the deadline** for a **penalty of 10%** of the marks associated with the deliverable item. **After 48 hours, a zero mark** will be assigned to the group without submission.
- Groups that design and implement a web-based, desktop, or mobile application user interface to support data-mining queries of the proposed solution and result visualization will be awarded up to **10% bonus marks**.
- The students are encouraged to contact the instructors by email and book an appointment to discuss the aspects of the projects and any other concerns.

For the project presentation assessment rubric, please go to the next page.

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Assessment Rubric					
Criteria	Level 4	Level 3	Level 2	Level 1	Score
Presentation of Ideas (10%)	2 points Student is able to express their idea clearly and answer the questions	1.5 points Student is able to express their ideas and answer most of the questions	1 point Student is somehow able to express their ideas and answer a few questions	0.5 point The student is not presenting their ideas well and is not able to answer most questions	/2
Design (20%)	4 points The approach to solving the problem is clearly described with relevant details	3 points The approach to solving the problem is clearly described with the most relevant details	2 points The approach to solving the problem is clearly described, but lacks some relevant details	1 point The approach to solving the problem is not clearly described, with minimal relevant details	/4
Demonstration of the solution/ functional requirements (40%)	8 points Introduce an innovative algorithm. Implementation is complete with additional features beyond the requirements. Demonstration works perfectly as intended	6 points Most functional requirements were modeled accurately. Implementation is complete with the intended solution. The demo works	4 points Some basic functional requirements were modeled. Implementation is complete; some of the project objectives are missing. The demo works with a few flaws	2 points The majority of functional requirements were missing, incomplete, and extremely inaccurate. Implementation is questionable; the demo is not working properly.	/8
Analysis of results and synthesis of information (30%)	6 points Uses in-depth analysis. Analysis is well presented and shows an excellent understanding of the project's results.	4.5 points Analysis is well presented and shows a good understanding of the project's results	3 points Analysis is presented and shows some understanding of the project's results	1.5 points Analysis is weak; the student does not understand the project's results	/6
Total =					/20